

Lifelong Learning for an Origami Robot: Supervised Learning of Target Adaptations

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I. INTRODUCTION

II. BACKGROUND

A. Lifelong Learning

B. Kresling Origami

III. METHODOLOGY

A. Target Adaptations

The method proposed in this paper aims to provide a way of improving a given controller's performance as well as fortifying it against evolving system dynamics *without* having access to the inner workings of the controller itself. In other words, provided that the most base assumptions of a controller hold true—i.e. it requires a state and a target as input and yields a control action—the method can be completely agnostic of the controller that is used.

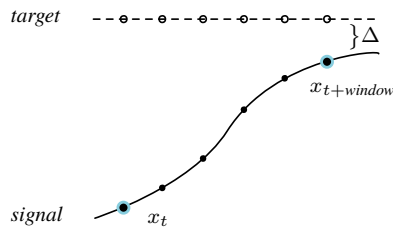


Fig. 1. The target adaptation problem: Based on the current x_t and the state some prediction window later $x_{t+window}$, the model learns to predict the target the controllers is trying to reach by representing it as a Δ to be added to $x_{t+window}$.

B. Toy Problem

To explore the proposed method, a toy problem is considered first. The toy problem consists of a simulation of a simple 1D mass-spring-damper system and an imperfectly tuned PID controller. With intervals of 5 seconds, a series of random

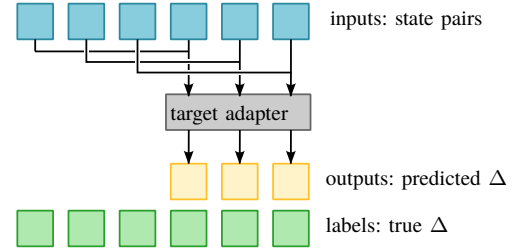


Fig. 2. The training process of the target adapter: recorded states spaced by a specified prediction window are paired up and provided as features to the adapter, which learns to predict a Δ , being the difference between the target and state $x_{t+window}$.

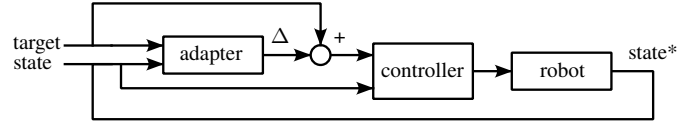


Fig. 3. Schematic overview of the full implementation: The adapter is used to offset the given target before yielding it to the controller, essentially *lying* to the controller in order to improve its performance.

targets are generated for the controller to reach and maintain. As the simulation progresses, the parameters of the dynamic system are slowly changed to simulate a degradation in the system.

C. Preliminary Results

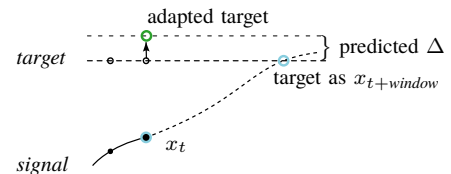


Fig. 4. Target adaptation during deployment: By supplying the model with the controller's target as $x_{t+window}$ it yields a Δ that can be added to the current target, which will supposedly result in the controller reaching the target within the specified window.

IV. A KRESLING ORIGAMI ROBOT

V. EXPERIMENTS AND RESULTS

VI. IMPROVED MODEL

VII. DISCUSSION